Constraints on spin-dependent exotic interactions between electrons at the nanometer scale

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New laboratory constraints are obtained on exotic interactions between electrons through pseudoscalar or pseudovector coupling by analyzing a recent measurement of the magnetic dipole-dipole interaction between two iron atoms at the nanometer scale. The analysis imposes the strongest bounds on the pseudovector coupling constant $(g_A^e)^2/4\pi\hbar c$ between electrons for exchange bosons heavier than 1 eV. Even with the counterpart of electron-positron interaction considered by assuming *CPT* invariance, the constraints are still the most stringent for bosons in the mass range from 1 eV to 20 eV. The constraints on the pseudoscalar coupling constant $(g_p^e)^2/4\pi\hbar c$ are only stronger than the limits derived from the trapped-strontium-ion experiment for exchange bosons heavier than 0.14 eV. Furthermore, a new experiment for more stringent constraints is proposed by measuring the interaction between one iron atom and multiple Holmium atoms.

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Extensions of the standard model predict particles which can mediate new interactions between fermions depending on their spins. A well-known example is axions which were initially proposed to explain the strong *CP* problem [1–3], and now they are promising candidates for dark matter [4–6]. Dipole-dipole interactions can arise from exchanging single axion or axionlike boson which couples to fermions via a pseudoscalar or derivative Lagrangian [7,8], and have a common effective interaction potential

$$V_P(r) = -\frac{g_p^2}{4\pi\hbar c}\frac{\hbar^3}{4m_f^2 c}\left[\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2\left(\frac{1}{\lambda r^2}+\frac{1}{r^3}\right) - (\boldsymbol{\sigma}_1\cdot\hat{\boldsymbol{r}})(\boldsymbol{\sigma}_2\cdot\hat{\boldsymbol{r}})\left(\frac{1}{\lambda^2 r}+\frac{3}{\lambda r^2}+\frac{3}{r^3}\right)\right]e^{-r/\lambda}. \quad (1)$$

Here $g_p^2/4\pi\hbar c$ is the dimensionless pseudoscalar coupling constant between the fermions, $\boldsymbol{\sigma}_{1,2}$ is the spin unit vector of the fermions, m_f is the fermion mass, r is the separation between the fermions, and $\lambda = \hbar/(m_b c)$ is the interaction range associated with exchange bosons of mass m_b. The pseudoscalar coupling constant is related to a symmetry-breaking scale in the derivative Lagrangian.

In general, Dobrescu and Mocioiu classified the interactions between two fermions by one-boson exchanging into sixteen types of potentials [9]. One of them has the same form of potential V_P. Another type of potential, which can be constrained by current analysis, is given by

$$V_A(r) = \frac{g_A^2}{4\pi\hbar c}\frac{\hbar c}{r}e^{-r/\lambda}\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2. \quad (2)$$

The interaction can be mediated by spin-1 bosons through pseudovector coupling, and $g_A^2/4\pi\hbar c$ is the corresponding dimensionless coupling constant.

In this paper, we will derive new constraints on the above exotic interactions between electrons based on a recent measurement of the magnetic dipole-dipole (MDD) interaction between two iron (Fe) atoms [10]. There have been many laboratory searches for the exotic spin-spin interactions between electrons and nucleons [11–19]. Up to now, stellar cooling rates imposed rather stronger constraints than laboratory experiments on the pseudoscalar coupling (g_p^e) to electrons [20–22]. However, these astrophysical bounds may suffer from various uncertainties [23]. The most stringent laboratory constraints on g_p^e and g_A^e for $m_b \leq 1$ meV were derived from the spin-polarized torsion balance experiments [16,17]. For m_b around the eV scale, the stringent constraints on g_p^e were derived from the measurement of the MDD interaction between two trapped strontium ions (Sr^+) at a separation of 2.4 μm [19]. The comparison between the hyperfine spectroscopy of positronium (Ps) and QED calculations sets limits on the exotic interactions between positron and electron [19,24,25].

Recently, the MDD interaction between two Fe atoms was measured by single-atom electron spin resonance (ESR) with scanning tunneling microscope (STM). The details of the experiment are referred to Refs. [10,26]. In summary, individual Fe atoms were deposited on bilayer magnesium oxide (MgO) films grown on a silver (Ag) substrate. The Fe atoms adsorb on atop oxygen on MgO,

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which results in a strong easy-axis magnetic anisotropy perpendicular to the surface plane. Electron spin resonance occurs between the two lowest-energy quantum spin states of the Fe atom, which are oriented in opposite out-of-plane directions (z) and split by an out-of-plane magnetic field (B_z). The resonance was driven by a radio frequency (rf) electric field applied between the STM tip and the Fe atom. By sweeping the rf-field frequency, ESR spectroscopy was measured by monitoring the spin-dependent tunneling current. The energy gap between the two states can be measured by the ESR frequency f_0 as $\Delta E = hf_0$.

This technique was used to measure the MDD interaction between two Fe atoms. With another target Fe atom presented nearby, the effective magnetic field ($B_z^{\rm eff}$) at the sensor Fe atom is modified due to the MDD interaction (see Fig. 2(c) of Ref. [10]). The $B_z^{\rm eff}$ depends on the magnetic state of the target Fe atom. If the magnetic moment of the target atom points up (down), $B_z^{\rm eff}$ is smaller (larger) than B_z and then the ESR frequency $f_\uparrow$ ($f_\downarrow$) is smaller (larger) than f_0. The ESR frequency splitting ($\Delta f = f_\downarrow - f_\uparrow$) is then related to the magnetic dipole-dipole potential energy as

$$\Delta f_{dd}(r) = \frac{4}{h}\frac{\mu_0}{4\pi}\frac{1}{r^3}[(m_z^{(1)}m_z^{(2)})], \tag{3}$$

where μ_0 is the vacuum permeability, r is the separation between the atoms, $m_z^{(i)}$ is the out-of-plane magnetic moment of atom i. For Fe-Fe interaction, we have $m_z^{(1)} = m_z^{(2)} = m_z$.

With the capability of atom manipulation by STM, the distance dependence of the interaction has been measured at the nanometer range [10]. The experimental data were found to agree with the MDD interaction very well. A single-parameter fit to Eq. (3) gives $m_z = 5.44 \pm 0.03\mu_B$, which is close to $5.2\mu_B$ determined by multiplet simulations of x-ray magnetic circular dichroism (XMCD) spectra [27]. The systematic errors have been thoroughly analyzed in the supplementary information of Ref. [10]. Control experiments have been performed with different magnetic fields, temperature and the structure of the tip apex. The results demonstrated that the ESR frequency splitting does not depend on these parameters for the range measured. They also discussed the validity of using Eq. (3) in case of considering the canting of the moment away from the surface normal, and found the relative error $\delta m_z/m_z$ is 0.001 due to the in-plane canting.

Constraints on the V_A-type interaction: The potentials V_P and V_A could change the ESR frequency similar to the MDD interaction. In contrast to the MDD interaction, here only the interactions between electron spins are involved in the energy splitting of the two lowest-energy quantum spin states. For Fe atom on MgO/Ag(100) substrate, the magnetic states were investigated by XMCD spectroscopy and scanning tunneling spectroscopy [27]. The multiplet calculations of XMCD spectra lead to that the two lowest-energy states are

$$|0\rangle = |L_z = -1.25, S_z = -1.96\rangle,$$
$$|1\rangle = |L_z = +1.25, S_z = +1.96\rangle. \tag{4}$$

The spin magnetic moment is $2\langle S_z\rangle\mu_B = 3.92\mu_B$, which is close to $4\mu_B$ for a free Fe atom and $3.7\mu_B$ as predicted by the density functional theory calculations [27]. The total electron spin of the Fe atom is predominately along the z direction, and its orientation is opposite for $|0\rangle$ and $|1\rangle$. Therefore, the potential given by Eq. (2) can be approximated to

$$V_A(r) = \frac{(g_A^e)^2}{4\pi\hbar c}\frac{4\hbar c}{r}e^{-r/\lambda}S_{z,1}S_{z,2}. \tag{5}$$

The interaction induces an extra frequency splitting

$$\Delta f_A(r) = \frac{(g_A^e)^2}{4\pi\hbar c}\frac{8c}{\pi r}\langle S_z\rangle^2 e^{-r/\lambda} \tag{6}$$

in the ESR spectroscopy.

Assuming that the ESR frequency splitting is contributed from both the MDD interaction and the pseudovector-mediated interaction, we fit the sum of $\Delta f_{dd}(r)$ and $\Delta f_A(r)$ to the data with two fitting parameters, m_z and $\alpha_A = (g_A^e)^2/4\pi\hbar c$, for different values of the fixed parameter λ. An example of such fits is shown in Fig. 1. As the two types of interactions have different distance dependence, the

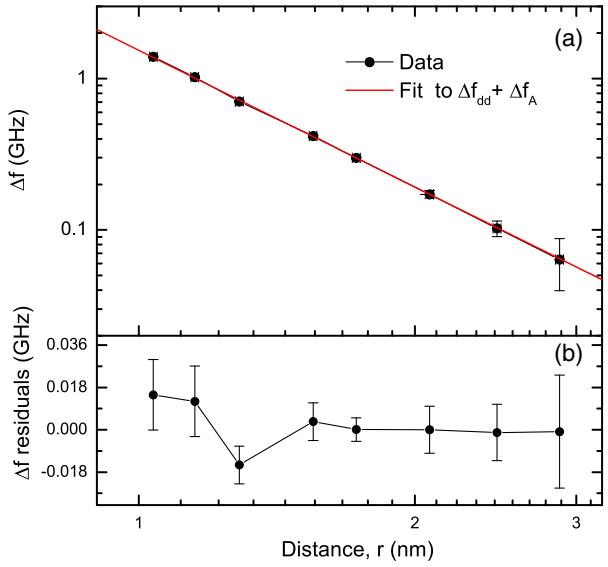


FIG. 1. (a) A plot of the ESR frequency splitting for different Fe-Fe distances (data points) with data selected from Ref. [10] for distance larger than 1 nm. An example of fit (red line), for $\lambda =$ 0.680 nm (corresponding to $m_b = 290$ eV), to the sum of Δf_{dd} and Δf_A gives $m_z = 5.44 \pm 0.42\mu_B$ and $\alpha_A = (0 \pm 3.3) \times 10^{-10}$. (b) The Δf residuals of the fit.

fitted curves are mainly contributed from the MDD interaction for the interaction range discussed in this work. The including of $\Delta f_A(r)$ in the fit leads to a larger uncertainty of m_z. The reduced χ^2 is obtained to be 0.887, slightly larger than the value (0.760) of a single-parameter fit to the MDD interaction. Within 2σ uncertainty, we find no evidence of the exotic interaction. We then derive a constraint at 95% confidential level by a sum of the fitted α_A and its 2σ uncertainty. The results are plotted in Fig. 2 with other constraints derived from the trapped-Sr^+-ion experiment [19] and Ps hyperfine spectroscopy [19,24,25]. The peak appears at $m_b \sim 290$ eV is due to an increase in the possible cancellation between the MDD interaction and the pseudovector-mediated interaction. The above analysis sets stronger constraints on the pseudovector coupling constant than those from the trapped-Sr^+-ion experiment for $m_b > 1$ eV. The constraints are also stronger than the positron-electron counterpart derived from Ps hyperfine spectroscopy for $m_b < 20$ eV. For $m_b < 1$ eV, the pseudovector-mediated interaction is still constrained by the trapped-Sr^+-ion experiment.

Constraints on the V_P-type interaction: The experimental data can also be used to constrain the V_P-type interaction. For interactions between the electrons of two Fe atoms, the potential V_P can be rewritten as

$$V_P(r) = -\frac{(g_p^e)^2}{4\pi\hbar c}\frac{\hbar^3}{m_e^2 c} S_{z,1} S_{z,2}\left(\frac{1}{\lambda r^2} + \frac{1}{r^3}\right) e^{-r/\lambda}. \quad (7)$$

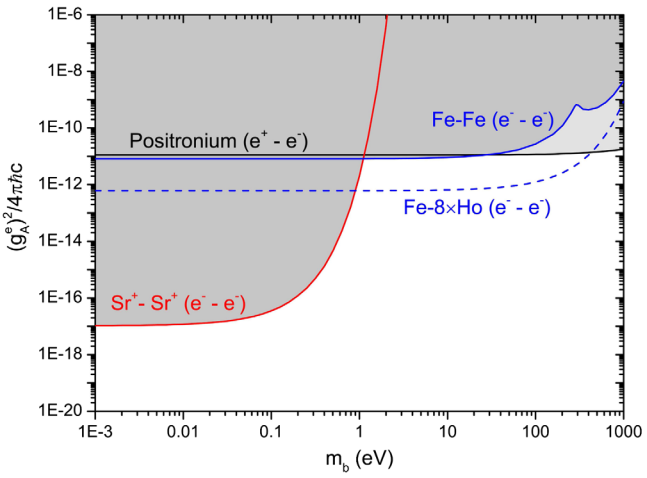


FIG. 2. Constraints on the dimensionless coupling constant, $(g_A^e)^2/(4\pi\hbar c)$, as a function of the boson mass m_b. The grey region is the parameter space excluded by the experiments for electron-electron interactions. The light grey region is the extra area excluded by the Ps spectroscopy experiment for positron-electron interactions. Fe-Fe: this work, Sr^+-Sr^+: trapped-Sr^+-ion experiment [19], Positronium: Ps hyperfine spectroscopy [19,24,25]. The dashed line shows the prospective limits of the proposed experiment.

Its contribution to the ESR frequency splitting is given by

$$\Delta f_P(r) = -\frac{(g_p^e)^2}{4\pi\hbar c}\frac{2\hbar^2}{\pi m_e^2 c}\langle S_z\rangle^2\left(\frac{1}{\lambda r^2} + \frac{1}{r^3}\right) e^{-r/\lambda}. \quad (8)$$

In the limit of λ going to infinity, the potential as well as Δf_P decays with distance as r^{-3}, therefore the exotic interaction is indistinguishable from the MDD interaction. We then perform a two-parameter fit of the sum of $\Delta f_{dd}(r)$ and $\Delta f_P(r)$ to the data only for $\lambda < 1.97$ nm (corresponding to $m_b > 100$ eV), and then derive the constraints as was done to the pseudovector-mediated interaction. Take a fit with $\lambda = 0.680$ nm for example, the fit results in $m_z = 5.44 \pm 0.07\mu_B$ and $\alpha_P = (g_p^e)^2/4\pi\hbar c = (0 \pm 8.2) \times 10^{-4}$ with a reduced χ^2 of 0.887.

For larger λ, the constraints on the exotic interaction can be obtained according to the following arguments. We note that the ESR frequency splitting contributed from the pseudoscalar-mediated interaction has an opposite sign to that of the MDD interaction, which instead has the same sign as the experimental data according to the definition of Δf. Therefore, we have $|\Delta f_P| \leq (\Delta f_{dd}^{\max} - \Delta f_{\exp})$, where $\Delta f_{dd}^{\max}$ is the upper limit of Δf_{dd}, and $\Delta f_{\exp}$ represents the measured value. The magnitude of Δf_{dd} is determined by m_z which reaches its maximum for a free Fe atom. A free Fe atom has an electronic configuration of $4s^2 3d^6$. It has six $3d$ outer-shell electrons and four of them are unpaired. According to Hund's rules, it has a total magnetic moment of $6\mu_B$. To obtain conservative limits on the coupling constant, we then fit the sum of $\Delta f_{dd}(r, m_z = 6\mu_B)$ and $\Delta f_P(r)$ to the data with single fitting parameter α_P for different values of λ. The constraint is then calculated by summing the fitted α_P with its 2σ uncertainty. As expected, the fitted curve is almost identical to what presented in Fig. 1. A fit for $\lambda = 197$ nm ($m_b = 1$ eV) leads to $\alpha_P = (3.04 \pm 0.07) \times 10^{-3}$ with a reduced χ^2 of 0.760.

The new constraints on the strength of pseudoscalar-mediated interactions are presented in Fig. 3. The constraints are stronger than those derived from the trapped-Sr^+-ion experiment for $m_b > 0.14$ eV, but are weaker than the limits on positron-electron exotic interactions derived from Ps hyperfine spectroscopy. However, it should be noted that the potentials V_P and V_A are even under the parity (P) and time-reversal (T) transformation. Only when we assumed CPT invariance, have electrons and positrons the same coupling magnitude for the pseudoscalar- and pseudovector-mediated interactions, and then the constraints on electron-electron and electron-positron interactions are comparable. However, there are also many possibilities that CPT invariance is violated [28]. In those cases, the results of electron-electron and electron-positron interactions should be treated as independent measurement.

Prospects for further improvements: The ESR-STM could be a useful technique to search for spin-dependent exotic interactions at the nanometer scale. The current

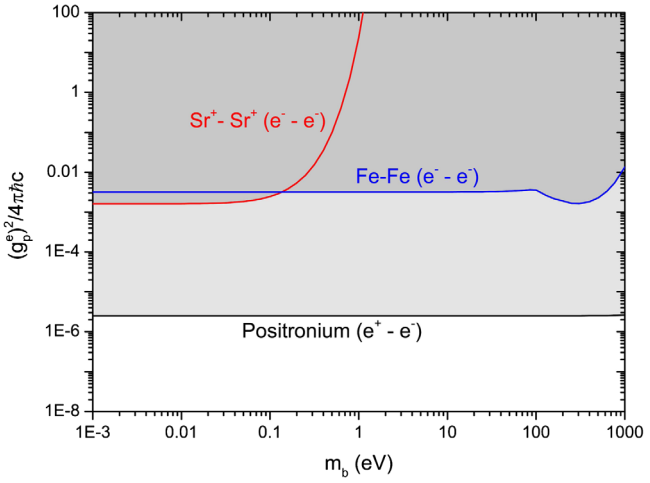


FIG. 3. Same as Fig. 2, for constraining the dimensionless pseudoscalar coupling constant $(g_p^e)^2/(4\pi\hbar c)$.

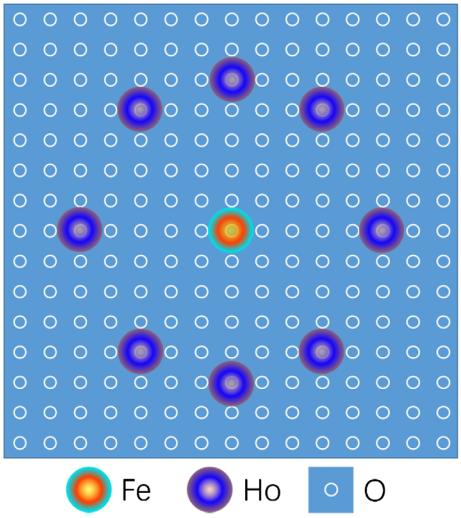


FIG. 4. Proposed atom configuration with eight Ho atoms surrounding one Fe sensor atom. The distance between the Ho atom and the Fe atom is set to five times of the lattice constant of MgO.

precision is limited by the drift or vibration of the STM tip [10]. Except to improve the instrumental precision, the constraints may be further strengthened by measuring the interaction between one sensor Fe atom and multiple target atoms. A suitable candidate for the target atom is the Holmium (Ho) atom as it was demonstrated to have a long magnetic state lifetime up to several hours [29]. Similar to Fe atom, the Ho atom on MgO/Ag(100) surface has strong out-of-plane magnetic anisotropy, which makes its electron spins orient in the out-of-plane direction through spin-orbital coupling. The measured magnetic moment of $10.1\mu_B$ suggests a $4f^{10}$ ion configuration [29] and thus the total electron spin should be 2 approximately. Furthermore, its magnetic state can be manipulated with a pulsed current applying through the STM tip, so that we can set all Ho atoms in either spin-up or spin-down state and then measure the ESR frequency difference between the two states. An example configuration is shown in Fig. 4 where eight Ho atoms are placed at the same distance from the center Fe sensor atom. The resultant ESR frequency difference is then magnified by approximately eight times comparing to a single pair of atoms. Assuming that the pseudovector coupling is constrained by current Δf measurement uncertainty of 5 MHz, we obtain the potential constraints as presented in Fig. 2. A main challenge for such experiments is to maintain the same distance between the STM tip and the sensor Fe atom during spin-up and spin-down measurements, in between of them the tip is needed to be moved away from the Fe atom to reverse the Ho magnetic state. For the pseudoscalar coupling, the magnetic dipole moment of Ho atoms cannot be measured independently, so that no stronger constraints can be obtained in this method.

In summary, we analyzed a recent measurement on the MDD interaction between two Fe atoms and derived new laboratory constraints on the pseudoscalar and pseudovector coupling constants between electrons. For bosons in the mass range from 1 eV to 20 eV, we obtained the strongest constraints on the pseudovector-mediated interaction even assuming *CPT* invariance. This analysis shows that the ESR-STM technique could be a useful technique to search for the spin-dependent exotic interactions at the nanometer scale. New experiment with further improvement was also proposed by measuring the interaction between one Fe sensor atom and eight Ho atoms.

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